

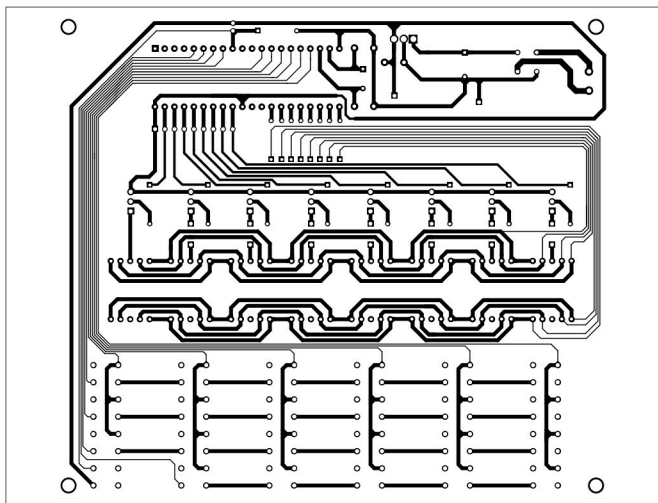


Fig. 3: An actual-size PCB layout for the celestial weight calculator

to make the PCB design simpler, as shown in Table IV.

The codes to display different numbers on even and odd 7-segment displays are shown in Table V.

The working of this device is very simple. Switch on the power supply and the device will display '0.' Enter

your weight on earth (either integer or real number). After entering weight press key for any celestial body and see the weight at that celestial body.

TABLE V
SSD Codes for Even and Odd SSDs

To see on SSD	Codes for even SSDs	Codes for odd SSDs
0	11H	88H
1	07H	BEH
2	32H	C4H
3	92H	94H
4	D4H	B2H
5	98H	91H
6	18H	81H
7	03H	BCH
8	10H	80H
9	90H	90H
BLANK	FFH	FFH

To enter a new weight on earth, press Reset key.

Software

The source program for the microcontroller is written in C language and compiled using Keil μ Vision4 compiler. The generated hex code is burnt

TABLE IV
Port Pin Connections with SSD Segments

Port pin	Even SSD	Odd SSD
P2.7	E	H(DP)
P2.6	D	C
P2.5	C	D
P2.4	H(DP)	E
P2.3	B	G
P2.2	A	F
P2.1	F	A
P2.0	G	B

TABLE VI
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
S6 pressed	DIS1 shows 0

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com